सहायक प्राध्यापक (दंतपरिवेष्टनशास्त्र, दंतविकृती व अणुजीवशास्त्र, मुखशल्यचिकित्साशास्त्र)
शासकीय दंत महाविद्यालय व रुग्णालय, वैद्यकिय शिक्षण व संशोधन संचालनालय, गट-ब
वैद्यकिय शिक्षण व औषधी द्रव्ये विभाग
Assistant Professor (Periodontology, Oral Pathology and Microbiology, Oral and Maxillofacial Surgery) Government Dental College and Hospital, Directorate of Medical Education and Research, Group-B
Medical Education and Drugs Department

परीक्षेचे टप्पे:- १) लेखी परीक्षा- २०० गुण **२)** मुलाखत - ५० गुण

-: परीक्षा योजना :-

विषय व सांकेतांक (११९२)	माध्यम	प्रश्नसंख्या	गुण	कालावधी	दर्जा	प्रश्नपत्रिकेचे स्वरुप
विभाग-अ विषयाशी/विभागाशी संबंधित घटक	इंग्रजी	५०	१००	एक तास	पदवी	वस्तुनिष्ठ बहुपर्यायी
विभाग-ब दंतपरिवेष्टनशास्त्र		५०	१००		पदव्युत्तर पदवी	
विभाग-क दंतविकृती व अणुजीवशास्त्र		५०	१००			
विभाग-ड मुखशल्यचिकित्साशास्त्र		५०	१००			

अ) नकारात्मक गुणदान -

१) प्रत्येक चुकीच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/ कमी करण्यात येतील.
२) एखाद्या प्रश्नाची एकापेक्षा अधिक उत्तरे दिली असल्यास अथवा ज्या उमेदवाराने उत्तरपत्रिकेत पूर्ण वर्तुळ चिन्हांकित केले नसेल अशा प्रश्नाचे उत्तर चुकीचे समजण्यात येऊन त्या प्रश्नाच्या उत्तराकरीता २५% किंवा १/४ एवढे गुण एकूण गुणांमधून वजा/कमी करण्यात येतील.
३) वरीलप्रमाणे कार्यपध्दतीचा अवलंब करताना एकूण अंतिम गुणांची बेरीज अपूर्णांकात आली तरीही ती अपूर्णांकातच राहील व पुढील कार्यवाही त्याच्या आधारे करण्यात येईल.
४) एखाद्या प्रश्नाचे उत्तर अनुत्तरित असेल तर, अशा प्रकरणी नकारात्मक गुणांची पध्दत लागू असणार नाही.

ब) अंतिम गुणवत्ता यादी ही वस्तुनिष्ठ परीक्षेतील व मुलाखतीतील एकत्रित गुणांवर आधारीत राहील.

क) दंतपरिवेष्टनशास्त्र विषयाकरीता विभाग-अ व विभाग-ब, दंतविकृती व अणुजीवशास्त्र विषयाकरीता विभाग-अ व विभाग-क व मुखशल्यचिकित्साशास्त्र विषयाकरीता विभाग-अ व विभाग-ड याप्रमाणे अभ्यासक्रम विहित करण्यात आला आहे.

-: अभ्यासक्रम :-

विषयाशी/विभागाशी संबंधित घटक यामध्ये खालील घटक व उपघटकांचा समावेश असेल.

Sr. No.	Topics and sub-topics
1	**SECTION A**
	HUMAN ANATOMY, EMBRYOLOGY, HISTOLOGY AND MEDICAL GENETICS I. INTRODUCTION II. HEAD and NECK: III.THORAX : IV. ABDOMEN V. CLINICAL PROCEDURES : VI. EMBRYOLOGY : VII. HISTOLOGY : VIII. MEDICAL GENETICS :

HUMAN PHYSIOLOGY

1. GENERAL PHYSIOLOGY
2. BLOOD:
3. MUSCLE AND NERVE
4. DIGESTIVE SYSTEM :
5. EXCRETORY SYSTEM :
6. BODY TEMPERATURE AND FUNCTIONS OF SKIN
7. ENDOCRINOLOGY
8. REPRODUCTION
9. CARDIO VASCULAR SYSTEM
10. RESPIRATORY SYSTEM
11. CENTRAL NERVOUS SYSTEM
12. SPECIAL SENSES

BIO-CHEMISTRY

1. CHEMISTRY OF BIO-ORGANIC MOLECULES
2. MACRONUTERIENTS AND DIGESTION
3. MICRONUTRIENTS
4. ENERGY METABOLISM
5. SPECIAL ASPECTS OF METABOLISM
6. BIOCHEMICAL GENETICS AND PROTEIN SYNTHESIS
7. ENZYME AND METABOLIC REGULATION
8. STRUCTURAL COMPONENTS AND BLOOD PROTEINS
9. MEDICAL BIOCHEMISTRY

GENERAL PATHOLOGY

1. Introduction to Pathology Terminologies
2. Etiology and Pathogenesis of Disease Cell Injury
3. Degenerations
4. Cell death and Necrosis Apoptasis
5. Inflammation
6. Healing
7. Tuberculosis
8. Syphilis
9. Typhoid
10. Thrombosis
11. Embolism
12. Ischaemia and Infraction
13. Derangements of body fluids
14. Disorders of circulation
15. Nutritional Disorders
16. Immunological mechanisms in disease
17. AIDS and Hepatitis.
18. Hypertension
19. Diabetes Mellitus
20. Adaptive disorders of growth
21. General Aspects of neoplasia
22 Anaemias
23. Leukaemias
24. Diseases of Lymph nodes
25. Diseases of oral cavity
26. Diseases of salivary glands
27. Common diseases of Bones
28. Diseases of Cardiovascular system
29. Haemorrhagic Disorders Coagulation cascade Coagulation disorders

A MICROBIOLOGY

B IMMUNOLOGY:
C SYSTEMATIC BACTERIOLOGY
D VIROLOGY
E MYCOLOGY
F PARASITOLOGY

GENERAL AND DENTAL PHARMACOLOGY AND THERAPEUTICS

General principles of pharmacology;
CNS drugs; General anaesthetics, hypnotics, analgescis psychotropic drugs, anti –
Autonomic drugs
Cardiovascular drugs
Autocoids
Drugs acting on blood
G.I.T. Drugs
Endocrines
Chemotherapy
Vitamins
Pharmacotherapy of emergencies in dental office
Chealating agents

GENERAL MEDICINE

1. Aims of medicine
2. Infections.
3. G.I.T.
4. CVS
5. RS
6. Hematology
7. Renal System Acute nephritis Nephrotic syndrome
8. Nutrition Avitaminosis
9. CNS
10. Endocrines
11. Critical care

GENERAL SURGERY

1. HISTORY OF SURGERY:
2. GENERAL PRINCIPLES OF SURGERY:
3. WOUNDS:
4. INFLAMMATION:
5. INFECTIONS:
6. TRANSMISSABLE VIRAL INFECTIONS:
7. SHOCK AND HAEMORRHAGE:
8. TUMOURS, ULCERS, CYSTS, SINUS AND FISTULAE:
9. DISEASES OF LYMPHATIC SYSTEM:
10. DISEASES OF THE ORAL CAVITY:
11. DISEASES OF LARYNX, NASOPHARYNX:
12. NERVOUS SYSTEM:
13. FRACTURES:
14. PRINCIPLES OF OPERATIVE SURGERY:
15. ANOMOLIES OF DEVELOPMENT OF FACE:
16. DISEASES OF THYROID AND PARATHYROID:
17. SWELLINGS OF THE JAW:
18. BIOPSY:

RESEARCH METHODOLOGY AND DENTAL EDUCATION TECHNOLOGY

1. Health Information: - Basic knowledge of Computers, MS Office, Window 2000, Statistical Programmes
2. Research Methodology: -Definition, types of research, designing a written proto

	Bio-Statistics: - Introduction, collection of data, presentation of data, Measures of Central tendency, measures of dispersion, Tests of significance, Sampling and sampling techniques- types, errors, bias, blind trails and calibration 3.Innovative Learning and Teaching Methodologies.	
2	**SECTION B (Periodontology)**	
1. Introduction: Definition of Periodontology, Periodontics, Periodontia, Brief historical background, Scope of Periodontics 2. Development of perio-dontal tissues, micro-structural anatomy and biology of periodontal tissues in detail Gingiva. Junctional epithelium in detail, Epithelial-Mesenchymal interaction, Periodontal, ligament Cementum, Alveolar bone. 3. Defensive mechanisms in the oral cavity: Role of-Epithelium,Gingival fluid, Saliva and otherdefensive mechanisms in the oral environment		
4. Age changes in periodontal structures and their significance in Geriatric dentistry	Age changes in teeth and periodontal structures and their association with periodontal diseases	
5. Classification of periodontal diseases	Need for classification, Scientific basis of Classification Classification of gingival and periodontal diseases as describedin World Workshop 1989 Gingivitis: Plaque associated, ANUG, steroid hormone influenced, Medication influenced, Desquamative gingivitis, other forms of gingivitis as in nutritional deficiency, bacterial and viralinfections etc. Periodontitis: Adult periodontitis, Rapidly progressive periodontitis A and B, Juvenile periodontitis (localized, generalized, and post-juvenile), Prepubertal periodontitis, Refractory periodontitis	
6. Gingival diseases	Localized and generalized gingivitis, Papillary, marginal and diffuse gingivitis Etiology, pathogenesis, Clinical signs, symptoms and management of i) Plaque associated gingivitis ii) Systemically aggravated gingivitis (sex hormones,drugs and systemic diseases) iii) ANUG iv) Desquamative gingivitis - Gingivitis associated with lichen planus, pemphigoid, pemphigus, and other vesiculobullous lesions v) Allergic gingivitis vi) Infective gingivitis-Herpetic, bacterial and candidial vii) Pericoronitis viii) Gingival enlargement (classification and differential diagnosis)	
7. Epidemiology of periodontal diseases	Definition of index, incidence prevalence, epidemiology, endemic, epidemic, and pandemic Classification of indices(Irreversible and reversible) Deficiencies of earlier indices used in Periodontics Detailed understanding of Silness and Loe Plaque Index ,Loe and Silness Gingival Index, CPITN and CPI. Prevalence of periodontal diseases in India and other countries. Public health significance (All these topics are covered at	

	length under community dentistry. Hence, the topics may be discussed briefly. However, questions may be asked from the topics for examination
8. Extension of inflammation from gingiva	Mechanism of spread of inflammation from gingival area to 1 deeper periodontal structures Factors that modify the spread
9. Pocket	Definition, signs and symptoms, classification, pathogenesis.
10. Etiology	- Dental Plaque (Biofilm) - Definition, New concept of biofilm - Types, composition, bacterial colonization, growth,maturation and disclosing agents - Role of dental plaque in periodontal diseases - Plaque microorganisms in detail and bacteria associated with periodontal diseases - Plaque retentive factors - Materia alba - Food debris - Calculus - Definition - Types, composition, attachment, theories of formation - Role of calculus in disease Food Impaction - Definition - Types, Etiology - Hirschfelds' classification - Signs ,symptoms and sequelae of treatment Trauma from occlusion - Definition, Types - Histopathological changes - Role in periodontal disease - Measures of management in briefHabits - Their periodontal significance - Bruxism and parafunctional habits, tongue thrusting, lipbiting, occupational habits IATROGENIC FACTORS Conservative Dentistry - Restorations - Contact point, marginal ridge, surface roughness, overhanging restorations, interface between restoration and teeth Prosthodontics - Interrelationship - Bridges and other prosthesis, pontics (types), surface contour, relationships of margins to the periodontium, Gingival protection theory, muscle action theory and theory of access to oral hygiene. Orthodontics - Interrelationship, removable appliances and fixed appliances - Retention of plaque, bacterial changesSystemic diseases - Diabetes, sex hormones, nutrition(Vit.C and proteins) - AIDS and periodontium

	Hemorrhagic diseases, Leukemia, clotting factor disorders, PMN disorders
11. Risk factors	Definition. Risk factors for periodontal diseases
12. Host response	Mechanism of initiation and progression of periodontal diseases Basic concepts about cells, Mast cells, neutrophils, macrophages, lymphocytes, immunoglobulins, complement system, immune mechanisms and cytokines in brief Stages in gingivitis-Initial, early, established and advanced Periodontal disease activity, continuous paradigm, random burst and asynchronous multiple burst hypothesis
13. Periodontitis	Etiology, histopathology, clinical signs and symptoms, diagnosis and treatment of adult periodontitis Periodontal abscess; definition,classification, pathogenesis,differential diagnosis and treatment Furcation involvement,Glickmans'classification, prognosis and management Rapidly progressive periodontitis Juvenile periodontitis: Localized and generalized Post-juvenile periodontitis Periodontitis associated with systemic diseases Refractory periodontitis
14. Diagnosis	Routine procedures, methods of probing, types of probes, (According to case history) Halitosis: Etiology and treatment. Mention advanced diagnostic aids and their role in brief
15. Prognosis	Definition, types, purpose and factors to be taken into consideration
16. Treatment plan	Factors to be considered
17. Periodontal therapy	A. General principles of periodontal therapy. Phase I,II, III, IV therapy. Definition of periodontal regeneration, repair, new attachment and reattachment. B. Plaque control i. Mechanical tooth brushes, interdental cleaning aids, dentifrices ii. Chemical; classification and mechanism of action of each and pocket irrigation
18. Pocket eradication procedures	Scaling and root planing: Indications Aims and objectives Healing following root planning Hand instruments, sonic, ultrasonic and piezo-electric scalers Curettage and present concepts Definition Indications Aims and objectives Procedures and healing response Flap surgery Definition Types of flaps, Design of flaps, papilla preservation Indications and contraindications

	- Armamentarium Surgical procedure and healing response
19. Osseous Surgery	Osseous defects in periodontal disease Definition Classification Surgery: resective, additive osseous surgery(osseous grafts with classification of grafts) Healing responses Other regenerative procedures; root conditioning Guided tissue regeneration
20. Mucogingival surgery and periodontal plastic surgeries	Definition Mucogingival problems: etiology, classification of gingival recession (P.D.Miller Jr. and Sullivan and Atkins) Indications and objectives Gingival extension procedures: lateral pedicle graft, frenectomy, frenotomy Crown lengthening procedures Periodontal microsurgery in brief
21. Splints	Periodontal splints Purpose and classification Principles of splinting
22. Hypersensitivity	Causes, Theories and management
23. Implants	Definition, types, scope and biomaterials used. Periodontal considerations: such as implant-bone interface, implant-gingiva interface, implant failure, peri-implantitis and management
24. Maintenance phase(SPT)	Aims, objectives, and principles Importance Procedures Maintenance of implants
25. Pharmaco-therapy	Periodontal dressings Antibiotics and anti-inflammatory drugs Local drug delivery systems
26. Periodontal management of medically compromised patients	Topics concerning periodontal management of medically compromised patients
27. Inter-disciplinary care	Pulpo-periodontal involvement Routes of spread of infection Simons' classification Management
28. Systemic effects of periodontal diseases in brief	Cardiovascular diseases, Low birth weight babies etc.
29. Infection control protocol	Sterilization and various aseptic procedures
30. Ethics	
3	**SECTION C (Oral Pathology and Microbiology)**

1. INTRODUCTION:

◆ A bird's eye view of the different pathological processes involving the oral cavity and oral cavity involvement in systemic diseases to be brought out. Interrelationship between General Medicine and General Surgery and Oral pathology to be emphasized.

1. Developmental disturbances of teeth, jaws and soft tissues of oral and paraoral region :

◆ Introduction to developmental disturbances - Hereditary, Familial mutation, Hormonal etc. causes to be highlighted.

◆ Developmental disturbances of teeth - Etiopathogenesis, clinical features, radiological features and histopathological features as appropriate :-

The size, shape, number, structure and eruption of teeth and clinical significance of the anomalies to be emphasized.

- Forensic Odontology.
- Developmental disturbances of jaws - size and shape of the jaws.
- Developmental disturbances of oral and paraoral soft tissues - lip and palate - clefts, tongue, gingiva, mouth, salivary glands and face.

2. Dental Caries :

- Etiopathogenesis, microbiology, clinical features, diagnosis, histopathology, immunology, prevention of dental caries and its sequelae.

3. Pulp and Periapical Pathology and Osteomyelitis.

- Etiopathogenesis and interrelationship, clinical features, microbiology, histopathology and radiological features (as appropriate) of pulp and periapical lesions and osteomyelitis.
- Sequelae of periapical abscess - summary of space infections, systemic complications and significance.

4. Periodontal Diseases :

- Etiopathogenesis, microbiology, clinical features, histopathology and radiological features (as appropriate) of gingivitis, gingival enlargements and periodontitis. Basic immunological mechanisms of periodontal disease to be highlighted.

5. Microbial infections of oral soft tissues :

- Microbiology, defence mechanisms including immunological aspects, oral manifestations, histopathology and laboratory diagnosis of common bacterial, viral and fungal infections namely :- Bacterial : Tuberculosis, Syphilis, ANUG and its complications - Cancrum Oris.

  Viral : Herpes Simplex, Varicella zoster, Measles, Mumps and HIV infection. Fungal : Candidal infection. Apthous Ulcers.

6. Common non- inflammatory diseases involving the jaws :

- Etiopathogenesis, clinical features, radiological and laboratory values in diagnosis of : Fibrous dysplasia, Cherubism, Osteogenesis Imperfecta, Paget's disease, Cleidocranial dysplasia, Rickets, Achondroplasia, Marfan's syndrome and Down's syndrome.

7. Diseases of TM Joint :

- Ankylosis, summary of different types of arthritis and other developmental malformations, traumatic injuries and myofascial pain dysfunction syndrome.

8. Cysts of the Oral and Paraoral region :

- Classification, etiopathogenesis, clinical features, histopathology, laboratory and radiological features (as appropriate) of Odontogenic cysts, Non-Odontogenic cysts, Pseudocysts of jaws and soft tissue cysts of oral and paraoral region.

9. Tumours of the Oral Cavity :

- Classification of Odontogenic, Non-Odontogenic and Salivary Gland Tumours. Etiopathogenesis, clinical features, histopathology, radiological features and laboratory diagnosis (as appropriate) of the following common tumours :-
  a) Odontogenic - all lesions.
  b) Non-odontogenic
  - Benign Epithelial - Papilloma, Keratoacanthoma and Naevi.
  - Benign Mesenchymal - Fibroma, Aggressive fibrous lesions, Lipoma, Haemangioma, Lymphangioma, Neurofibroma, Schwannoma, Chondroma, Osteoma and Tori.
  - Malignant Epithelial - Basal Cell Carcinoma, Verrucous Carcinoma, Squamous Cell carcinoma and Malignant Melanoma
  - Malignant Mesenchymal - Fibrosarcoma, Osteosarcoma, Giant cell tumour, Chondrosarcoma, Angiosarcoma,

Kaposi's sarcoma, Lymphomas,
Ewing's sarcoma and Other
Reticuloendothelial tumours.

c) Salivary Gland

- Benign Epithelial neoplasms - Pleomorphic Adenoma, Warthin's tumour and Oncocytoma.

Malignant Epithelial neoplasms –
Adenoid Cystic Carcinoma,
Mucoepidermoid Carcinoma,
Acinic Cell Carcinoma and Adenocarcinomas.

d) Tumours of Disputed Origin - Congenital Epulis and Granular Cell Myoblastoma.

e) Metastatic tumours - Tumors metastasizing to and from oral cavity and the routes of metastasis.

10. Traumatic, Reactive and Regressive lesions of Oral Cavity :

- Pyogenic and Giant cell granuloma, exostoses Fibrous Hyperplasia, Traumatic Ulcer and Traumatic Neuroma.
- Attrition, Abrasion, Erosion, Bruxism, Hypercementosis, Dentinal changes, Pulp calcifications and Resorption of teeth.
- Radiation effects of oral cavity, summary of Physical and Chemical injuries including allergic reactions of the oral cavity.
- Healing of Oral wounds and complications - Dry socket.

11. Non neoplastic Salivary Gland Diseases :

- Sialolithiasis, Sialosis, Sialadenitis, Xerostomia and Ptyalism.

12. Systemic Diseases involving Oral cavity :

- Brief review and oral manifestations, diagnosis and significance of common Blood, Nutritional, Hormonaland Metabolic diseases of Oral cavity.

13. Mucocutaneous Lesions :

- Etiopathogenesis, clinical features and histopathology of the following common lesions. Lichen Planus, Lupus Erythematosus, Pemphigus and Pemphigoid lesions, Erythema Multiforme, Psoriasis, Scleroderma, Ectodermal Dysplasia, Epidermolysis bullosa and White sponge nevus.

14. Diseases of the Nerves :

- Facial neuralgias - Trigeminal and Glossopharyngeal. VII nerve paralysis, Causalgia.
- Psychogenic facial pain and Burning mouth syndrome.

15. Pigmentation of Oral and Paraoral region and Discolouration of teeth :

- causes and clinical manifestations.

16. Diseases of Maxillary Sinus :

- Traumatic injuries to sinus, Sinusitis, Cysts and Tumours involving antrum.

17. a) ORAL PRECANCER – CANCER; Epidemiology, aetiology, clinical and histopatholotgical features, TNM classification. Recent advances in diagnosis, management and prevention.

b) Biopsy : Types of biopsy, value of biopsy, cytology, histo chemistry and frozen sections in diagnosis of oral diseases.

18. Principles of Basic Forensic Odontology (Pre-clinical Forensic Odontology):

- Introduction, definition, aims and scope.
- Sex and ethnic (racial) differences in tooth morphology and histological age estimation
- Determination of sex and blood groups from buccal mucosa / saliva.
- Dental DNA methods
- Bite marks, rugae patterns and lip prints.
- Dental importance of poisons and corrosives.

Overview of forensic medicine and toxicology

4	SECTION D (Oral and Maxillofacial Surgery)

1. Introduction, definition, scope, aims and objectives.
2. Diagnosis in oral surgery:
   (A) History taking
   (B) Clinical examination
   (C) Investigations.
3. Principles of infection control and cross-infection control with particular reference to HIV/AIDS and Hepatitis.
4. Principles of Oral Surgery -
   a) Asepsis: Definition, measures to prevent introduction of infection during surgery.
      1. Preparation of the patient
      2. Measures to be taken by operator
      3. Sterilisation of instruments - various methods of sterilisation etc.
      4. Surgery set up.
   b) Painless Surgery:
      1. Pre-anaesthetic considerations. Pre-medication: purpose, drugs used
      2. Anaesthetic considerations -
         a) Local b) Local with IV sedations
      3. Use of general anaesthetic
   c) Access:
      Intra-oral: Mucoperiosteal flaps, principles, commonly used intra oral incisions. Bone Removal: Methods of bone removal.
      Use of Burs: Advantages and precautions
      Bone cutting instruments: Principles of using chisel and osteotome.
      Extra-oral: Skin incisions - principles, various extra-oral incision to expose facial skeleton.
         a) Submandibular
         b) Pre auricular
         c) Incision to expose maxilla and orbit
         d) Bicoronal incision
   d) Control of haemorrhage during surgery Normal Haemostasis
      Local measures available to control bleeding Hypotensive anaesthesia etc.
   e) Drainage and Debridement
      Purpose of drainage in surgical wounds
      Types of drains used
      Debridement: purpose, soft tissue and bone debridement.
   f) Closure of wounds
      Suturing: Principles, suture material, classification, body response to various materials etc.
   g) Post operative care
      Post operative instructions Physiology of cold and heat Control of pain - analgesics Control of infection - antibiotics
      Control of swelling - anti-inflammatory drugs Long term post operative follow up - significance.
5. Exodontia: General considerations Ideal Extraction.
   Indications for extraction of teeth
   Extractions in medically compromised patients. Methods of extraction -
   (a) Forceps or intra-alveolar or closed method. Principles, types of movement, force etc.
   (b) Trans-alveolar, surgical or open method, Indications, surgical procedure.
   Dental elevators: uses, classification, principles in the use of elevators, commonly used elevators.
   Complications of Exodontia - Complications during exodontia Common to both maxilla and mandible. Post-operative complications
   Prevention and management of complications.

6. Impacted teeth:
   Incidence, definition, aetiology.
   (a) Impacted mandibular third molar. Classification, reasons for removal, Assessment - both clinical and radiological Surgical procedures for removal. Complications during and after removal, Prevention and management.
   (b) Maxillary third molar,
   Indications for removal, classification, Surgical procedure for removal.
   (c) Impacted maxillary canine Reasons for canine impaction,
   Localization, indications for removal,
   Methods of management, labial and palatal approach, Surgical exposure, transplantation, removal etc.
7. Pre-prosthetic Surgery:
   Definition, classification of procedures
   (a) Corrective procedures: Alveoloplasty, Reduction of maxillary tuberosities, Frenoctemies and removal of tori.
   (b) Ridge extension or Sulcus extension procedures Indications and various surgical procedures
   (c) Ridge augmentation and reconstruction. Indications, use of bone grafts, Hydroxyapatite Implants - concept of osseo integration Knowledge of various types of implants and surgical procedure to place implants.
8. Diseases of the maxillary sinus Surgical anatomy of the sinus. Sinusitis both acute and chronic
   Surgical approach of sinus - Caldwell-Luc procedureRemoval of root from the sinus.
   Oro-antral fistula - aetiology, clinical features and various surgical methods for closure.
9. Disorders of T.M. Joint
   Applied surgical anatomy of the joint.
   Dislocation - Types, aetiology, clinical features and management. Ankylosis - Definition, aetiology, clinical features and management
   Myo-facial pain dysfunction syndrome, aetiology, clinical features, management- Non surgical and surgical.
   Internal derangement of the joint. Arthritis of T.M. Joint.
10. Infections of the Oral cavity
   Introduction, factors responsible for infection, course of odontogenic infections, spread of odontogenic infections through various facial spaces. Dento-alveolar abscess - aetiology, clinical features and management.
   Osteomyelitis of the jaws - definition, aetiology, pre-disposing factors, classification, clinical features and management.
   Ludwigs angina - definition, aetiology, clinical features, management and complications.
11. Benign cystic lesions of the jaws - Definition, classification, pathogenesis.
   Diagnosis - Clinical features, radiological, aspiration biopsy, use of contrast media and histopathology.
   Management - Types of surgical procedures, Rationale of the techniques, indications, procedures, complications etc.
12. Tumours of the Oral cavity -General considerations
   Non odontogenetic benign tumours occuring in oral cavity - fibroma, papilloma, lipoma, ossifying fibroma, mynoma etc.
   Ameloblastoma - Clinical features, radiological appearance and methods of management.
   Carcinoma of the oral cavity -Biopsy - types
   TNM classification.
   Outline of management of squamous
   Cell carcinoma: surgery, radiation and chemotherapy

Role of dental surgeons in the prevention and early detection of oral cancer.

13. Fractures of the jaws -

General considerations, types of fractures, aetiology, clinical features and general principles of management.

mandibular fractures - Applied anatomy, classification. Diagnosis - Clinical and radiological

Management - Reduction closed and open Fixation and immobilisation methods

Outline of rigid and semi-rigid internal fixation.

Fractures of the condyle - aetiology, classification, clinical features, principles of management.

Fractures of the middle third of the face.

Definition of the mid face, applied surgical anatomy, classification, clinical features and outline of management.

Alveolar fractures - methods of management Fractures of the Zygomatic complex

Classification, clinical features, indications for treatment, various methods of reduction and fixation.

Complications of fractures - delayed union, non-union and malunion.

14. Salivary gland diseases -

Diagnosis of salivary gland diseases' Sialography, contrast media, procedure. Infections of the salivary glands

Sialolithiasis - Sub mandibular duct and gland and parotid duct. Clinical features, management.

Salivary fistulae

Common tumours of salivary glands like Pleomorphic adenoma including minor salivary glands.

15. Jaw deformities -

Basic forms - Prognathism, Retrognathism and open bite. Reasons for correction.

Outline of surgical methods carried out on mandible and maxilla.

16. Neurological disorders -

Trigeminal neuralgia - definition, aetiology, clinical features and methods of management including surgical.

Facial paralysis - Aetiology, clinical features. Nerve injuries - Classification, neurorrhaphy etc.

17. Cleft Lip and Palate -

Aetiology of the clefts, incidence, classification, role of dental surgeon in the management of cleft patients. Outline of the closure procedures.

18. Medical Emergencies in dental practice –

Primary care of medical emergencies in dental practice particularly -

(a) Cardio vascular (b) Respiratory (c) Endocrine (d) Anaphylactic reaction (e) Epilepsy

19. Emergency drugs and Intra muscular I.V. Injections -

Applied anatomy, Ideal location for giving these injections, techniques etc.

20. Oral Implantology

21. Ethics

LOCAL ANAESTHESIA:

Introduction, concept of L.A., classification of local anaesthetic agents, ideal requirements, mode of action, types of local anaesthesia, complications.

Use of Vaso constrictors in local anaesthetic solution - Advantages, contra-indications, various vaso constrictors used. Anaesthesia of the mandible -

Pterygomandibular space - boundaries, contents etc. Interior Dental Nerve Block - various techniques Complications

Mental foramen nerve block Anaesthesia of Maxilla - Intra - orbital nerve block.

Posterior superior alveolar nerve block Maxillary nerve block - techniques.

GENERAL ANAESTHESIA –

Concept of general anaesthesia.
Indications of general anaesthesia in dentistry. Pre-anaesthetic evaluation of the patient. Pre-anaesthetic medication - advantages, drugs used. Commonly used anaesthetic agents.
Complication during and after G.A.
I.V. sedation with Diazepam and Medozolam. Indications, mode of action, technique etc. Cardiopulmonary resuscitation
Use of oxygen and emergency drugs. Tracheostomy.

Note: The above syllabus is as per the B.D.S. course regulation notified by the Dental Council of India.

दिनांक – ११ जून, २०२६

सरिता रि. बांदेकर-देशमुख
परीक्षा नियंत्रक व सह सचिव
महाराष्ट्र लोकसेवा आयोग